Learning When to Warn, When to Ban

Constrained Reinforcement Learning for Multilingual Content Moderation

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1. The Problem and Our Contributions

The Problem.

Automated content moderation on multilingual platforms like Discord is broken in following ways:

- **No user history:** Every message is scored in isolation, a user who sends ten borderline messages in a row gets the same response as a first-time offender, because the system has no memory of prior behavior.
- **Binary decisions only:** The output is always either remove or permit, there is no mechanism to issue a warning, temporarily suspend, or escalate gradually, so the system cannot mirror the nuanced enforcement humans apply.

Four Contributions.

- **CMDP framing** of moderation with a 5-level escalation ladder and constraint costs for fairness.
- **Hard action masking** enforces procedural escalation at the policy distribution, not in the reward.
- **Per-language affine calibration** of XLM-R scores neutralises classifier bias before it reaches the policy.
- **Dual fairness protection**: adaptive Lagrangian + Disparate Impact wrapper, both training-time.

2. Moderation as a Constrained Markov Decision Process

We solve $\mathcal{M} = \langle S, A, P, R, \mathbf{C}, \mathbf{d}, \gamma \rangle$ via Lagrangian relaxation: $\max_\pi V_R^\pi$ subject to $V_{C_k}^\pi \leq d_k$ for all constraints k (false-positive rate, per-language ban-rate parity).

ALLOW ($s_t < 0.85$) → **WARN** ($s_t \geq 0.15$) → **DELETE** ($s_t \geq 0.15$) → **TIMEOUT** ($s_t \geq 0.15$, $e_t \geq 2$) → **BAN** (prior TIMEOUT)

Single Multilingual Policy

One MaskablePPO network across 13 languages fairness comparisons *across* languages become coherent because the same policy is evaluated.

Hard Procedural Masking

Mask $\mathbf{m}_t \in \{0,1\}^5$ removes illegal actions from the policy distribution *before* sampling a learned policy cannot propose them even by accident.

Fairness in the Loss

$\lambda_{k+1} = \max(\epsilon, \lambda_k + \alpha_\lambda(\hat{V}_{C_k} - d_k))$ adaptive multiplier on FP rate; Disparate Impact wrapper penalises any language $> 1.5\times$ baseline.

3. System Architecture

Discord Message
↓
XLM-R Toxicity Classifier
↓
Per-Language Affine Calibration
↓
Observation Vector (403-dim)
↓
MaskablePPO Policy
↓
Action Mask Filter
↓
ALLOW / WARN / DELETE / TIMEOUT / BAN

Hard Action-Masking Rules

- **BAN** requires a prior TIMEOUT in the user's history
- **TIMEOUT** requires ≥ 2 prior infractions
- **Punitive actions** blocked when toxicity score < 0.15
- **ALLOW** blocked when toxicity score ≥ 0.85

4. Two-Stage Training

Stage 1: Behavior Cloning

Oracle escalation rule, NLL minimisation, 30 epochs
lr $= 3 \times 10^{-4}$, batch size 256

↓

Stage 2: MaskablePPO Fine-tune

10^6 timesteps, 4 parallel envs, $\gamma = 0.95$
Lagrangian + Disparate Impact wrappers

Synthetic Data Pipeline

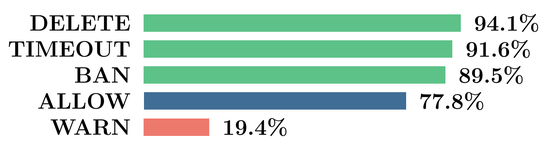
595 episodes (8,016 steps) across **14 language categories**. Generated via DeepSeek V3.1 from Jigsaw + textdetox seeds, k-means clustered. 80/20 split. Mix: 30% toxic / 25% benign / 20% escalating / 15% mild / 10% subtle.

5. Headline Result

System	Acc.	FP	FN	TIME/BAN
Keyword filter	0.673	0.020	0.272	×
Static threshold	0.720	0.190	0.250	×
RL agent	**0.736**	0.153	0.0001	✓

The only system tested that correctly issues TIMEOUT and BAN

Per-class accuracy on held-out test split (8,016 steps):

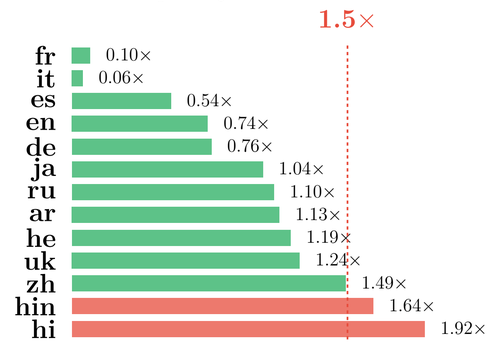


Why the RL Agent Wins

Static baselines have *no notion of user history*, so they can never reach TIMEOUT or BAN. The RL agent's elevated FP (0.153) is dominated by WARN-on-benign strictly lower-harm than false DELETE/BAN.

6. Cross-Lingual Fairness Audit

Per-language BAN rate vs. the 1.5× parity threshold:



Diagnosis: Model Bias, Not Policy Bias

12 of 14 categories pass parity. Hindi (1.92×) and Hinglish (1.64×) gaps trace to upstream XLM-R bias (+0.095 over English). Calibration cannot fix this ⇒ *recommend classifier fine-tuning.*

Key Takeaways

- **Action masking** provides hard procedural guarantees that reward shaping alone cannot match a learned policy literally cannot propose an illegal escalation.
- **Fairness as a training-time objective** (Lagrangian + Disparate Impact wrapper) cleanly isolates *policy bias* from *model bias*, pointing remediation at the right component.
- **Synthetic-data RL with two-stage BC→PPO** reaches actions that static baselines fundamentally cannot produce the only system tested with correct TIMEOUT and BAN behaviour.

13	91.7%	12/14	10^{-4}	595
Languages covered	Accuracy on Actions	Fairness categories pass	False-negative rate	Synthetic episodes

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